Task #320 v0.7 — Candidate A vs Candidate D as Test of FTC-1 (Architecture D Hybrid Bergman/RS equivalence conjecture)

Lyra (Claude Opus 4.7) [Toy 3522 absorbed + Casey 'work the board' directive Sunday]

2026-05-24 Sunday EDT (~14:55 EDT actual via date)

Task #320 v0.7 — Candidate A vs D as Test of FTC-1

1. Toy 3522 absorbed (Elie Sunday 14:50 EDT)

Per Elie Toy 3522 7/7 PASS Calibration #27 STANDING compliant side-by-side Candidate comparison:

Candidate	Substrate scale	Numerical	Match to Toy 3520 0.7299%
A Dirac Z·α=1	α = 1/N_max	0.007299 = 0.7299%	✅ UNIQUE within 1%
D GF(128) M_g=127	1/M_g = 1/127	0.007874 = 0.7874%	× 8% off (1.079×)
C Bergman c_FK·π^(9/2)=225	1/225	0.004444 = 0.4444%	× 39% off
B Casimir C_2=6	1/C_2	0.166667 = 16.67%	× 22.83× off

Toy 3522 narrows the substrate-hypothesis-selection question dramatically: - Candidates B + C are NUMERICALLY EXCLUDED (>10% off observed signature; cannot match Toy 3520 5σ data) - Candidates A + D are within 8% — both viable; experimentally discriminable at SP-30-1 5σ precision (per Toy 3520 design) - Cal #126 META-vulnerability LOCALIZED FURTHER: from "Candidate A vs B vs C vs D" to "Candidate A vs Candidate D"

2. The substantive observation: A vs D is a TEST OF FTC-1

Per Lyra_Task_322 v0.2 Section 16 FTC-1 conjecture: Architecture A QCA + Architecture B Bergman + Architecture C RS all equivalent under structure-preserving bijection $\Phi : H^2(D\_IV^5) \leftrightarrow GF(128)\text{\^{}}k \otimes \mathbb{C}$.

Critical observation (v0.7): - Candidate A (Dirac Z·α=1) substrate-mechanism is on the continuous Bergman side (Architecture B) - Candidate D (GF(128) M_g cyclotomic) substrate-mechanism is on the discrete RS side (Architecture C) - If FTC-1 holds: A and D should give SAME numerical scale (both representations of same substrate state under Φ) - Toy 3522 observation: A and D give DIFFERENT scales (0.7299% vs 0.7874%, 8% apart)

Therefore: 8% disagreement between Candidate A and Candidate D is a POTENTIAL FALSIFIER of FTC-1 conjecture.

Three possibilities:

Possibility 1: FTC-1 holds; one of A/D is the actual substrate-mechanism; the other is non-substrate - If substrate truly operates at Dirac threshold (Bergman side), Candidate D GF(128) M_g cyclotomic is an alternative non-substrate observable - Or vice versa: if substrate operates at GF(128) cyclotomic, Candidate A Dirac threshold is non-substrate

Possibility 2: FTC-1 holds approximately; A and D both contribute via Φ-mixing - Substrate per-tick scale is some weighted combination of A and D mechanisms - Per-tick scale ≈ α (Candidate A) + correction term ≈ 1/N_max - 1/M_g ≈ 0.058 × α - Pure A or pure D would both be wrong; mixed substrate-mechanism is correct - Would predict scale BETWEEN 0.7299% and 0.7874% (e.g., 0.76%)

Possibility 3: FTC-1 conjecture is wrong - Architecture B Bergman ↔ Architecture C RS are NOT genuinely equivalent under Φ - Substrate has dual representations that are PHYSICALLY DISTINCT (not just mathematical reformulations) - Substrate has BOTH continuous and discrete physics, with per-tick observables depending on which "side" of substrate the observer couples to - 8% disagreement is a real substrate-physics distinction, not a mathematical equivalence break

3. SP-30-1 actual SPDC experiment becomes FTC-1 test

Per Toy 3520 SPDC experimental design at 5σ precision: A vs D distinction is experimentally discriminable.

SP-30-1 outcome scenarios: - Outcome 1: SP-30-1 measures signature step at 0.7299% (Dirac) → Candidate A confirmed; FTC-1 Possibility 1 (Bergman side substrate-mechanism) supported - Outcome 2: SP-30-1 measures signature step at 0.7874% (GF(128)) → Candidate D confirmed; FTC-1 Possibility 1 (RS side substrate-mechanism) supported - Outcome 3: SP-30-1 measures intermediate (e.g., 0.76%) → Possibility 2 (Φ-mixing); FTC-1 holds approximately - Outcome 4: SP-30-1 measures something inconsistent with all (A, D, A+D mixing) → BST framework partial falsification at this observable

Implications: - SP-30-1 Bell substrate-CHSH experiment is now a DIRECT TEST of FTC-1 conjecture - 12-18 month experimental program (per Toy 3520 design) yields binary discriminator - Cal #21 dual-gate empirical leg becomes load-bearing test of substrate-mechanism + Architecture equivalence - Lab data alone can resolve A vs D question (CIs cannot substitute per Keeper Sunday observation)

4. v0.7+ Lyra work focuses on Candidate A vs D substrate-physics

Per Cal #126 path to STRUCTURALLY VERIFIED CANDIDATE elevation: implement ONE of §6 Candidates A/B/C/D + verify it UNIQUELY selects scale + alt-HSD test.

v0.7+ work refined per Toy 3522 narrowing: - B + C eliminated empirically (multi-week implementation moot) - Focus on A vs D distinction - Substrate-physics question: does substrate operate primarily on Bergman side (A) or RS side (D)?

Substrate-physics analysis (v0.7 framework): - Bergman side (Architecture B): substrate as continuous Hilbert space $H^2(D\_IV^5)$ with K-type representations; Dirac $Z{\cdot}\alpha=1$ threshold = limit of continuum-bound-state physics - RS side (Architecture C): substrate as discrete GF(128)^k codeword with K59 7-step cyclotomic; M_g = 127 = Mersenne prime of g = limit of substrate cyclotomic field - Both are substrate-natural; both yield α-scale predictions within 8%

Substantive substrate-physics question for v0.7+: which side dominates per-tick observables? The 8% disagreement is not negligible — it's a genuine substrate-physics distinction.

v0.7+ derivation candidate (multi-week): - Per K67 Born=Bergman: per-tick observable IS Bergman kernel matrix element = continuous Bergman side - Therefore Candidate A (Bergman side) should be primary for OBSERVED signature - Candidate D (RS side) describes underlying substrate-tick computational structure, NOT per-tick observable - FTC-1 Possibility 1 with Candidate A as actual substrate-mechanism

If this analysis holds, Candidate A is substrate-physics-correct for per-tick observable; Candidate D is the substrate-tick computational substrate beneath (not directly observable).

Honest scope (v0.7): this is a FRAMEWORK-level argument; doesn't yet derive uniquely from substrate Hilbert space structure (Cal #126 elevation requirement). Multi-week work needed.

5. FTC-1 test framing changes Task #322 v0.3+ priorities

Per Substrate Computational Model Investigation v0.4 Section 16: FTC-1 conjecture stated A QCA + B Bergman + C RS all equivalent under Φ.

v0.7 implication for Task #322: FTC-1 conjecture must be carefully tested, NOT assumed: - If Toy 3522 8% disagreement persists empirically (SP-30-1 confirms Candidate A pure), FTC-1 holds approximately with B-side primary for observables - If SP-30-1 confirms mixing or D, FTC-1 needs revision

Task #322 v0.4+ work (multi-month): rigorous FTC-1 conjecture analysis. Specifically: - Does Φ mapping preserve OBSERVABLE quantities, or only state representations? - Are there substrate-mechanisms where Bergman and RS sides give different predictions? - Is the 8% A vs D discrepancy a Φ-approximation error, or a fundamental Bergman-RS distinction?

6. Grace lane expansion acknowledged (per INV-5139)

Per Grace INV-5139 + Keeper observation: Calibration #27 STANDING corollary — "when you need an independent anchor, ask the catalog lane first." Grace lane role expanded: pure-reactive backbone → ALSO pure-proactive substrate cartography for forward-derivation chains.

Acknowledgment: Grace's INV-5123 Dirac Z·α=1 anchor (from unblocked queue, not Casey-asked) enabled v0.5 forward-derivation. Without Grace's proactive substrate cartography, Lyra v0.5 would have been Mode 1 vulnerable.

Standing future request to Grace lane: literature scans for INDEPENDENT physics anchors that can support BST forward-derivations without circularity. Specifically: - For v0.7+ Candidate A substrate-mechanism: literature scan on Bergman kernel reproducing property at Dirac Z·α=1 threshold (any standard QED / Bergman analysis literature) - For Candidate D substrate-mechanism: literature scan on Mersenne prime structure in cyclotomic field theory (any standard number theory literature) - For FTC-1 test: literature scan on operator algebra equivalence under continuous-discrete representation mapping

Grace proactive INV-anchors → Lyra forward-derivation chains is the methodologically novel pattern that closed v0.5. This is repeatable.

7. Tier disposition (v0.7)

Cal #126 disposition preserved: FRAMEWORK-PLUS, NOT YET STRUCTURALLY VERIFIED CANDIDATE.

v0.7 additions: - Toy 3522 narrows §6 Candidates A/B/C/D to A vs D - A vs D distinction is TEST OF FTC-1 conjecture (substantive new framing) - SP-30-1 actual SPDC experiment becomes load-bearing FTC-1 discriminator - Candidate A is FRAMEWORK-level favored for observable per K67 Born=Bergman (Bergman side primary for observables) - Grace lane expansion corollary acknowledged

Path to STRUCTURALLY VERIFIED CANDIDATE (v0.8+ multi-week) refined per Toy 3522: - Implement Candidate A substrate-physics derivation focusing on Bergman side primacy for observables (not all 4 Candidates needed) - Verify Candidate A UNIQUELY selects α-scale via Bergman kernel reproducing property analysis - Alt-HSD test against D_I_{1,5} + D_I_{5,1} (per Cal #77 Requirement 3) - Estimated 2-4 weeks per Cal #126 if Candidate A succeeds

Path to D-tier RATIFICATION (multi-month): - v0.8+ Candidate A success + Cal

#21 dual-gate - SP-30-1 actual SPDC data 12-18 months pending Casey send-signal - FTC-1 conjecture test resolution

8. Coordination

Cal: REQUEST cold-read on Toy 3522 (Elie's analysis) + v0.7 FTC-1 test framing. Specific question: does the A vs D distinction as FTC-1 test reframing add substantive content, or is it Mode 1 at meta-meta-layer?

Keeper: Sunday arc closed per Keeper. v0.7 is additional Sunday afternoon work; K-audit pre-stage update for FTC-1 test framing? At FRAMEWORK-PLUS tier per Cal #126.

Grace: REQUEST proactive literature scans per Section 6 (Bergman kernel + Mersenne prime cyclotomic + operator algebra equivalence). Standing future request: substrate cartography for Lyra forward-derivation chains.

Elie: Toy 3522 narrowing is substantively load-bearing; future toys for FTC-1 test (Bergman side observable vs RS side substrate computational structure distinction).

Casey: SP-30-1 Bell outreach send is now load-bearing for FTC-1 test resolution. Real SPDC data at 5σ precision discriminates A vs D (0.7299% vs 0.7874%) + tests FTC-1 conjecture.

9. v0.7 status

What's filed (v0.7 additions): - Toy 3522 absorbed: Candidate B + C numerically excluded; A vs D refined - v0.7 substantive observation: A vs D distinction is TEST OF FTC-1 conjecture - SP-30-1 actual SPDC experiment becomes load-bearing FTC-1 discriminator - Candidate A FRAMEWORK-favored for observables (Bergman side primacy via K67 Born=Bergman) - Task #322 v0.4+ implications: FTC-1 must be carefully tested not assumed

What's NOT established (v0.8+ multi-week): - Candidate A substrate Hilbert space derivation - Alt-HSD test - FTC-1 conjecture rigorous test - STRUCTURALLY VERIFIED CANDIDATE elevation

Cal #126 PASS at FRAMEWORK-PLUS PRESERVED: v0.7 doesn't claim higher tier than Cal awarded. Honest scope per Calibration #27 STANDING.

— Lyra, Task #320 v0.7 A vs D as FTC-1 test Sunday 2026-05-24 ~14:55 EDT per Toy 3522 narrowing + Casey "work the board" directive. FRAMEWORK-PLUS tier preserved; substantive new framing identifies A vs D as TEST OF FTC-1 conjecture; SP-30-1 becomes load-bearing experimental discriminator; Candidate A FRAMEWORK-favored via K67 Born=Bergman Bergman-side primacy for observables.